刘昳豪 应数23-2

21-22

一、均值 195.5 中位数 196.5 极差 124

二、$X\sim N(2,1)$，$X-2\sim N(0,1)$，$\frac{Y_i}{2}\sim N(0,1)$，$i=1,2,3,4$

$$T=\frac{4(X-2)}{\sqrt{\sum_{i=1}^{4}Y_i^2}}=\frac{(X-2)}{\sqrt{\frac{1}{4}\sum_{i=1}^{4}(\frac{Y_i}{2})^2}}\sim t(4)\text{，}P(|T|>t_0)=0.01\text{ 则 }P(|T|\le t_0)=0.99$$

$t_0=t_{0.995}(4)=4.604$

三、(1). $\bar{X}\sim N(\mu,\frac{\sigma^2}{n})$，$\underline{\frac{(n-1)S^2}{\sigma^2}\sim\chi^2(n-1)}$

$E(\bar{X})=\mu$，$D(\bar{X})=\frac{\sigma^2}{n}=E(\bar{X}^2)-[E(\bar{X})]^2$，$E(\bar{X}^2)=\mu^2+\frac{\sigma^2}{n}$

$\chi^2(n-1)$ 的期望

$\underline{E[\frac{(n-1)S^2}{\sigma^2}]=n-1}=\frac{n-1}{\sigma^2}E(S^2)$，$E(S^2)=\sigma^2$，$E(T)=E(\bar{X}^2)-\frac{1}{n}E(S^2)=\mu^2$

凑 $\chi^2(1)$

方差

(2). $\underline{D[\frac{(n-1)S^2}{\sigma^2}]=2n-2}=\frac{(n-1)^2}{\sigma^4}D(S^2)$，$D(S^2)=\frac{2\sigma^4}{n-1}$；$\underline{\bar{X}\sim N(0,\frac{1}{n})\text{，}\sqrt{n}\bar{X}\sim N(0,1)}$

$\therefore n\bar{X}^2\sim\chi^2(1)$，$E(n\bar{X}^2)=nE(\bar{X}^2)=1$，$E(\bar{X}^2)=\frac{1}{n}$；$D(n\bar{X}^2)=n^2D(\bar{X}^2)=2$，$D(\bar{X}^2)=\frac{2}{n^2}$

由 $\bar{X}$ 和 S^2 独立，$D(T)=D(\bar{X}^2)+\frac{1}{n^2}D(S^2)=\frac{2}{n^2}+\frac{1}{n^2}(\frac{2\sigma^4}{n-1})$ 这里忘记把1带入了

四、(1). $L(\theta)=\prod_{i=1}^{n}(\frac{1}{\theta}e^{-\frac{x_i}{\theta}})=\frac{1}{\theta^n}e^{-\frac{\sum x_i}{\theta}}$，记 $\sum x_i=n\bar{x}$

$\ln L=-n\ln\theta-\frac{n\bar{x}}{\theta}\triangleq\varphi(\theta)$，$\varphi'(\theta)=-\frac{n}{\theta}+\frac{n\bar{x}}{\theta^2}$，$\varphi'(\theta)=0$，$\theta=\bar{x}$

$\theta\in(0,\bar{x})$，$\varphi'(\theta)>0$；$\theta>\bar{x}$，$\varphi'(\theta)<0$，故 $L(\theta)_{max}=L(\bar{x})$，$\hat{\theta}=\bar{x}$

(2). $\prod_{i=1}^{n}(\frac{1}{\theta}e^{-\frac{x_i}{\theta}})=\frac{1}{\theta^n}e^{-\frac{n\bar{x}}{\theta}}$，~~g(x,θ)~~ $g(\bar{x},\theta)=\frac{1}{\theta^n}e^{-\frac{n\bar{x}}{\theta}}$，$h(x_1,\dots,x_n)=1$，是充分统计量

(3). $g(\theta)=\theta$，$g'(\theta)=1$

由各 x_i 独立，$E(\hat{\theta})=\frac{1}{n}\cdot n\theta=\theta$；$D(\hat{\theta})=\frac{1}{n^2}\cdot n\cdot\theta^2=\frac{\theta^2}{n}$

更是UMVUE

$I(\theta)=-E[\frac{\partial^2}{\partial\theta^2}\ln f]=-E(\frac{1}{\theta^2}-\frac{2}{\theta^3}X)=\frac{1}{\theta^2}$，方差下界：$\frac{[g'(\theta)]^2}{nI(\theta)}=\frac{\theta^2}{n}=D(\hat{\theta})$，故是有效估计

五、$H_0: p_{10}=p_{20}=\dots=p_{60}=\frac{1}{6}$

$$\chi^2=\frac{(24-20)^2}{20}+\frac{(12-20)^2}{20}+\frac{(19-20)^2}{20}+\frac{(21-20)^2}{20}+\frac{(16-20)^2}{20}+\frac{(28-20)^2}{20}\approx 8.1$$

$\chi^2_{0.95}(5)=11.071>8.1$　　接受 H_0，均匀

六 (1).

134.452	3	44.8513	15.51
47.72	16	2.3925	
182.172	19		

(2). σ^2 的最大似然：$\frac{47.72}{20}=2.236$　无偏：2.3925

(3). $15.51>F_{0.99}(3,16)=5.29$　显著（拒绝 H_0）

$H_0: \mu_1=\mu_2=\mu_3=\mu_4$

七. (1). $\hat{b} = \frac{l_{xy}}{l_{xx}} = \frac{1}{l_{xx}}[\sum x_i y_i - n\bar{x}\bar{y}] = \frac{1}{l_{xx}}[\sum x_i y_i - \bar{x}\sum y_i] = \frac{1}{l_{xx}}\sum(x_i-\bar{x})y_i$

$= \sum \frac{x_i-\bar{x}}{l_{xx}} y_i$ ，是无偏估计，因为

$E(\hat{b}) = \sum \frac{x_i-\bar{x}}{l_{xx}} E(y_i)$

$= \sum \frac{x_i-\bar{x}}{l_{xx}}(a+bx_i)$

$= \sum \frac{x_i-\bar{x}}{l_{xx}} a + \sum \frac{x_i-\bar{x}}{l_{xx}} x_i \cdot b$

$= 0 + b = b$

(2). $\hat{b} = \frac{985.5 - \frac{1}{15}\times 141.25\times 104.5}{1332.8125 - \frac{1}{15}(141.25)^2} \approx 0.67$ ，$\hat{a} = \bar{y} - \hat{b}\bar{x} \approx 0.66$

$\hat{y} = 0.66 + 0.67x$

(3). $S_T = 729.625 - 728.016 = 1.609$

$S_R = 0.67\times 0.67\times l_{xx} = 1.2$

$S_e = S_T - S_R = 0.409$

$F = \frac{S_R}{S_e/13} \approx 38.14$ ， $t = \frac{\hat{b}}{\hat{\sigma}/\sqrt{l_{xx}}} \approx 6.18$ ， $r = \frac{l_{xy}}{\sqrt{l_{xx}\cdot l_{yy}}} \approx 0.70$

$38.14 > F_{0.99}(3,16)$ ， $6.18 > t_{0.975}(13)$， $0.70 > \sqrt{\frac{F_{0.99}(3,16)}{F_{0.99}(3,16)+13}}$ 显著

八. （不太会这道题，但感觉挺像高代里的单射，感觉应该往这方面扯）

$X \sim p(x,\theta) = p(x,\theta(u))$ ，$\hat{\theta}$是θ的最大似然估计量，$\exists$唯一的 $\hat{u} = u(\hat{\theta})$ ，由单值反函数性质

$L_{max} = L(\hat{\theta}) = p(x_1, \hat{\theta}(\hat{u}))\dots p(x_n, \hat{\theta}(\hat{u}))$

$= p^*(x_1, \hat{u}(\hat{\theta}))\dots p^*(x_n, \hat{u}(\hat{\theta}))$

$= p^*(x_1, \hat{u})\dots p^*(x_n, \hat{u})$

$= L(\hat{u})$

22-23.

刘映豪 应数23-2

一. $X_{n+1} \sim N(\mu,\sigma^2)$, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$. $X_{n+1} - \bar{X} \sim N(0, \frac{n+1}{n}\sigma^2)$ （由X_{n+1}与$\bar{X}$独立）

$(X_{n+1} - \bar{X})/\sqrt{\frac{n+1}{n}\sigma^2} \sim N(0,1)$; $\frac{n\hat{\sigma}^2}{\sigma} \sim \chi^2(n-1)$ ，故

$V = \frac{(X_{n+1}-\bar{X})/\sqrt{\frac{n+1}{n}}\sigma}{\hat{\sigma}\sqrt{\frac{n}{n-1}}/\sigma} = \frac{(X_{n+1}-\bar{X})/\sqrt{\frac{n+1}{n}}}{\sqrt{\frac{n\hat{\sigma}^2}{n-1}}} \sim t(n-1)$

二. (1). $E(X) = \int_0^\theta x\cdot\frac{1}{\theta}dx = \frac{\theta}{2}$ ， $\frac{\hat{\theta}}{2} = \bar{x}$ ， $\hat{\theta} = 2\bar{x}$

(2). $L(\theta) = \frac{1}{\theta^n}$ ，要使 L 最大，只要 θ^n 最小，即 θ 最小，由 $0 < x_{(1)} \le x_{(2)} \le \dots \le x_{(n)} \le \theta$

$\hat{\theta}_M = X_{(n)}$

(3). $E(T_1) = \frac{2}{n}\cdot n\cdot\frac{\theta}{2} = \theta$ ； $X_{(n)}$的密度函数$p_n(x) = \frac{n!}{(n-1)!}(\frac{x}{\theta})^{n-1}\cdot\frac{1}{\theta} = \frac{n}{\theta^n}x^{n-1}$. $E(X_{(n)}) = \int_0^\theta \frac{n}{\theta^n}x^n dx = \frac{n}{n+1}\theta$

$\therefore E(T_2) = \frac{n+1}{n}\cdot(\frac{n}{n+1}\theta) = \theta$

$\boxed{2}$

三. (1). $n=16$, $\frac{(n-1)S^2}{\sigma^2}\sim\chi^2(15)$, $P(\frac{S^2}{\sigma^2}\le 2.0386)=P(\frac{15S^2}{\sigma^2}\le 30.579)=P(\chi^2(15)\le\chi^2_{0.99}(15))=0.99$

(2). $D[\frac{(n-1)S^2}{\sigma^2}]=2n-2$, $\therefore \frac{225}{\sigma^4}D(S^2)=30$, $D(S^2)=\frac{2}{15}\sigma^4$

四. $\bar{x}_1=11.4$, $\bar{x}_2=9.7$, $S_1^2=\frac{1.44+1.44+0.09+0.81+0.09+0.81+0.64+0.64}{8-1}\approx 0.85$,

$S_2^2=\frac{1.21+3.24+1+2.25+2.89+0.01+0.04+0.16+1.44+0.16}{10-1}\approx 1.38$; $S_w^2=\frac{7\times 0.85+9\times 1.38}{7+9}\approx 1.15$

$\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{S_w\sqrt{\frac{1}{m}+\frac{1}{n}}}\sim t(m+n-2)$, $m=8, n=10$, 置信区间为 $\bar{x}_1-\bar{x}_2\pm t_{0.975}(16)S_w\sqrt{\frac{1}{m}+\frac{1}{n}}$

$[0.544, 2.856]$

五. (1). $\alpha=P(X<48$ 或 $X>72\mid P=0.3)$

$X\sim b(200, P=0.3)$ $E(X)=60$, $D(X)=42$, 用二项分布中心极限定理

$=P(X<48\mid P=0.3)+P(X>72\mid P=0.3)$

$=P(\frac{X-60}{\sqrt{42}}<\frac{-12}{\sqrt{42}})+P(\frac{X-60}{\sqrt{42}}>\frac{12}{\sqrt{42}})$ 注意到 $\frac{12}{\sqrt{42}}=1.85$, $\alpha=[1-\Phi(1.85)]\times 2=0.0644$

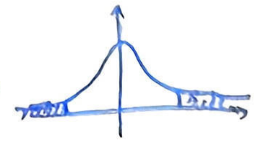
(2). $\beta=P(48<X<72\mid P=0.2)$

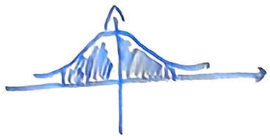
$X\sim b(200, 0.2)$

$=P(\frac{8}{\sqrt{32}}<\frac{X-40}{\sqrt{32}}<\frac{32}{\sqrt{32}})$

注意到 $\frac{8}{\sqrt{32}}\approx 1.41$, $\sqrt{32}\approx 5.66$

$=\Phi(5.66)-\Phi(1.41)\approx 0.0793$

(1)


(2)、


六. $H_0: p_{10}=\frac{1}{4}, p_{20}=\frac{1}{2}, p_{30}=\frac{1}{4}$

$\chi^2=\frac{(26-30)^2}{30}+\frac{(66-60)^2}{60}+\frac{(28-30)^2}{30}\approx 1.27<\chi^2_{0.95}(2)$ 接受H_0, 一致

七. (1).

1290	2	645	13.65
567	12	47.25	
1857	14		

(2)、 $13.65>F_{0.95}(2,12)$ 拒绝H_0, 有影响

八. (1). $l_{xx}=115.11-\frac{(30.3)^2}{9}=13.1$, $l_{xy}=345.09-\frac{30.3\times 91.1}{9}\approx 38.39$, $l_{yy}=1036.65-\frac{(91.1)^2}{9}\approx 114.52$

$\hat{\beta}=\frac{l_{xy}}{l_{xx}}\approx 2.93$, $\hat{\alpha}=\bar{y}-\hat{\beta}\bar{x}\approx 0.26$ $\hat{y}=2.93x+0.26$

(2). $S_T=l_{yy}=114.52$

$S_R=\hat{\beta}^2\cdot l_{xx}\approx 112.46$

$S_e=S_T-S_R=2.06$

$F=\frac{S_R}{S_e/(9-2)}\approx 382.15>F_{0.95}(1,7)=5.59$

$t=\frac{\hat{\beta}}{\hat{\sigma}/\sqrt{l_{xx}}}\approx 19.59>t_{0.975}(7)=2.3646$

$r=\frac{l_{xy}}{\sqrt{l_{xx}l_{yy}}}\approx 0.70>\sqrt{\frac{F_{0.95}(1,7)}{F_{0.95}(1,7)+7}}$ 显著

23-24

一、由 X_1、X_2 独立，$X_1+X_2 \sim N(0,2\sigma^2)$，$X_1-X_2 \sim N(0,2\sigma^2)$

若 X_1+X_2 与 X_1-X_2 独立，则 $Y=\dfrac{(X_1+X_2)^2/2\sigma^2}{(X_1-X_2)^2/2\sigma^2} \sim F(1,1)$. 下证 X_1+X_2 与 X_1-X_2 独立.

由 X_1+X_2 与 X_1-X_2 均为正态分布，独立与不相关等价，只要证 $Cov(X_1+X_2, X_1-X_2)=0$

$$\begin{aligned}Cov(X_1+X_2, X_1-X_2) &= Cov(X_1,X_1)-Cov(X_1,X_2)+Cov(X_1,X_2)-Cov(X_2,X_2)\\ &= D(X_1)-D(X_2)\\ &= 0\end{aligned}$$

课本P167. 不相关不一定独立，但独立一定不相关；但在二维正态场合下，独立$\Leftrightarrow$不相关.

二. $P(X_1,X_2,\dots,X_n)=\theta^n\cdot(1-\theta)^{x_1+x_2+\dots+x_n}=\theta^n(1-\theta)^t$. $g(T,\theta)=\theta^n(1-\theta)^t$，$h(x_1,x_2,\dots,x_n)=1$

$P(X_1,X_2,\dots,X_n)=\theta^n(1-\theta)^t=g(T,\theta)\cdot h(x_1,\dots,x_n)$ T是充分统计量.

三、(1). $p(x)=\frac{1}{\theta}$，$\theta<x\le 2\theta$. $E(X)=\int_\theta^{2\theta}\frac{x}{\theta}dx=\frac{3}{2}\theta$. $\frac{3}{2}\hat\theta_1=\bar x$，$\hat\theta_1=\frac{2}{3}\bar x$ 是 θ 的矩估计

$L(\theta)=(\frac{1}{\theta})^n$. $\theta\le x_{(1)}\le x_{(2)}\le\dots\le x_{(n)}\le 2\theta$. 要使 $L(\theta)$ 最大，只要让 θ 最小

$\frac{x_{(n)}}{2}\le\theta<x_{(1)}$，故 $\hat\theta_2=\frac{1}{2}x_{(n)}$ 为 θ 的最大似然估计

(2). 对 $\hat\theta_1$：$E(\hat\theta_1)=\frac{2}{3}\cdot\frac{1}{n}\cdot n\cdot\frac{3}{2}\theta=\theta$ $\quad E(X^2)=\int_\theta^{2\theta}\frac{x^2}{\theta}dx=\frac{7}{3}\theta^2$. $D(X)=E(X^2)-[E(X)]^2=\frac{1}{12}\theta^2$

$D(\hat\theta_1)=\frac{4}{9}\cdot\frac{1}{n^2}\cdot n\cdot D(X)=\frac{1}{27n}\theta^2\to 0\ (n\to\infty)$. 由于 $E(\hat\theta_1)=\theta$，$D(\hat\theta_1)\to 0\ (n\to\infty)$

$\hat\theta_1$ 是 θ 的无偏估计、相合估计

对 $\hat\theta_2$：$p(x)=\frac{1}{\theta}$，$0<x\le 2\theta$；$F(x)=\frac{x-\theta}{\theta}$，$0<x\le 2\theta$. $p_n(x)=n\cdot(\frac{x-\theta}{\theta})^{n-1}\cdot\frac{1}{\theta}=\frac{n}{\theta^n}(x-\theta)^{n-1}$，$\theta<x\le 2\theta$

$E(\hat\theta_2)=\frac{1}{2}\int_\theta^{2\theta}\frac{nx}{\theta^n}(x-\theta)^{n-1}dx=\frac{2n+1}{2(n+1)}\theta$，$E(\hat\theta_2^2)=\frac{1}{4}\int_\theta^{2\theta}\frac{nx^2}{\theta^n}(x-\theta)^{n-1}d\theta=\theta^2\left[\frac{n}{n+1}-\frac{1}{2(n+1)(n+2)}\right]$

但 $\lim_{n\to\infty}E(\hat\theta_2)=\theta$

$D(\hat\theta_2)=E(\hat\theta_2^2)-[E(\hat\theta_2)]^2=\theta^2\left[-(\frac{2n+1}{2n+2})^2+\frac{n}{n+1}-\frac{1}{(n+1)(n+2)}\right]\to 0\ (n\to\infty)$ 无偏× 相合✓

四. (1). 由 $\frac{\bar x-\mu}{s/\sqrt{n}}\sim t(n-1)$，$n=15$，$\alpha=0.05$，置信区间为 $[\bar x-t_{0.975}(15)\frac{s}{\sqrt{15}}, \bar x+t_{0.975}(15)\frac{s}{\sqrt{15}}]$

即 [9.78, 10.22]

(2). 检验统计量 $t=\frac{\bar x-\mu}{s/\sqrt{n}}$ 3，拒绝域 $W:\{t \mid |t|\ge t_{0.975}(15)=2.1314\}$

拒绝H_0

五、$P(t)=\begin{cases}\frac{1}{24}, & t\in[0,24] \\ 0 & 其它\end{cases}$, $P_{10}=P_{20}=\dots=P_{120}=\int_0^2 \frac{1}{24}dt=\frac{1}{12}$

$H_0: P_{10}=\dots=P_{120}=\frac{1}{12}$

$\chi^2=\frac{(266-240)^2}{240}+\frac{(281-240)^2}{240}+\dots+\frac{(266-240)^2}{240}=\frac{1}{240}[676+1681+81+4+64+225+961+961+1936+676+441]$

$=\frac{1}{240}\times 7706$

$\approx 32.11 > \chi^2_{0.95}(11)$

拒绝H_0，不服从均匀分布

六、$g(\lambda)=\lambda$, $g'(\lambda)=1$

$E(\bar{X})=\frac{1}{n}\cdot n\cdot E(X)=\lambda$ ，$\bar{X}$是λ的无偏估计

$D(\bar{X})=\frac{1}{n^2}\cdot n\cdot D(X)=\frac{\lambda}{n}$ ， $\ln P=x\ln\lambda-\ln x!-\lambda$ ， $\frac{\partial \ln P}{\partial\lambda}=\frac{x}{\lambda}-1$ ， $\frac{\partial^2\ln P}{\partial\lambda^2}=-\frac{x}{\lambda^2}$

$I(\theta)=-E(\frac{\partial^2}{\partial\lambda^2}\ln P)=E(\frac{x}{\lambda^2})=\frac{1}{\lambda}$ 　方差下界：$\frac{[g'(\lambda)]^2}{nI(\theta)}=\frac{\lambda}{n}=D(\bar{X})$ ，$\bar{X}$是λ的有效估计

七、(1)

443.6	3	147.87	2.15
1513.515	22	68.80	
1957.115	25		

(2) σ^2的无偏估计：68.8；最大似然估计：58.21

(3) $2.15 < F_{0.95}(3,22)=3.05$，接受$H_0$，无显著差异

八、(1) $Q(k)=\Sigma(y_i-kx_i)^2$ ，$Q'(k)=\Sigma -2x_i(y_i-kx_i)=-2\Sigma(x_iy_i-kx_i^2)$ ，$Q'(k)=0$ ，$k=\frac{\Sigma x_iy_i}{\Sigma x_i^2}$

$k\in(-\infty,\frac{\Sigma x_iy_i}{\Sigma x_i^2})$ 时 $Q'(k)<0$；$k>\frac{\Sigma x_iy_i}{\Sigma x_i^2}$ 时 $Q'(k)>0$. 故 $\hat{k}=\frac{\Sigma x_iy_i}{\Sigma x_i^2}$

估计值：$\hat{k}=\frac{3.03+4.512+6.034+8.032+9.918+12.5+14.828+22.204+27.72}{9.4864+14.1376+18.5761+25.2004+30.3601+39.6025+45.4276+54.76+72.9316+85.3776}$

把模型换了，和课本的不一样

$\hat{k}\approx 0.275$

这里应该算错了
不愿意再按一遍计算器了，感觉没什么意义
一位有耐心的同学算的是0.320

(1) 由课本P_{398}改写
(2) 由课本P_{400} 第三个式子改写

(2) 由 $y=kx+\varepsilon$, $\varepsilon\sim N(0,\sigma^2)$, $y_i\sim N(kx_i,\sigma^2)$.

$E(\hat{k})=\frac{\Sigma x_iE(y_i)}{\Sigma x_i^2}=\frac{\Sigma x_i^2\cdot k}{\Sigma x_i^2}=k$ 是

拓展：由课本P_{400}，可以证该模型下 $D(\hat{k})=\frac{\sigma^2}{\Sigma x_i^2}$ ，及 $\hat{k}\sim N(k,\frac{\sigma^2}{\Sigma x_i^2})$ ，比课本上的模型简单，但你一定要反应过来这个模型没有"β_0"与课本上不同.

九、①寻找一个好的点估计$\hat{\theta}$ ②构造枢轴量$G(\hat{\theta},\theta)$，含待估参数θ和估计量$\hat{\theta}$，分布已知，不含任何参数

③由置信水平$1-\alpha$确定a、b，使 $P(a\le G(\hat{\theta},\theta)\le b)=1-\alpha$

④等价变形为 $P(\hat{\theta}_L\le\theta\le\hat{\theta}_U)=1-\alpha$ 　⑤$[\hat{\theta}_L,\hat{\theta}_U]$为$\theta$的置信水平$1-\alpha$的置信区间.

2024~2025 A

一、$T^2 \sim \chi^2(n)$

设 $X_1 \sim N(0,1)$，$X_2 \sim \chi^2(n)$，X_1、X_2 相互独立，$T = \frac{X_1}{\sqrt{X_2/n}} \sim t(n)$.

则 $T^2 = \frac{X_1^2}{X_2/n}$，其中 $X_1^2 \sim \chi^2(1)$，$X_2 \sim \chi^2(n)$，X_1^2 与 X_2 独立．故 $T^2 \sim F(1,n)$

二．$E(X) = \int_0^1 \theta x^\theta dx = \frac{\theta x^{\theta+1}}{\theta+1}\Big|_0^1 = \frac{\theta}{\theta+1} \triangleq \eta$　　$\cancel{D(X)}$ $E(X^2)$ $= \int_0^1 \theta x^{\theta+1} dx = \frac{\theta}{\theta+2} x^{\theta+2}\Big|_0^1 = \frac{\theta}{\theta+2}$

θ 的矩估计 $\hat{\theta}$：令 $\frac{\hat{\theta}}{\hat{\theta}+1} = \bar{X}$，则 $\hat{\theta} = \frac{1}{1-\bar{X}} - 1$　　$D(X) = E(X^2) - [E(X)]^2 = \theta\left[\frac{1}{\theta+2} - \frac{\theta}{(\theta+1)^2}\right]$

相合性：$\bar{X}$ 是 η 的相合估计，因为 $E(\bar{X}) = \frac{1}{n}[E(X_1) + \dots + E(X_n)] = \frac{1}{n}\cdot n \cdot \eta = \eta$

$D(\bar{X}) = \frac{1}{n^2}[D(X_1) + \dots + D(X_n)] = \frac{1}{n}\cdot \cancel{\frac{\theta}{\theta+2}} \to 0\ (n \to \infty)$

$\frac{1}{n}\cdot\theta\left[\frac{1}{\theta+2} - \frac{\theta}{(\theta+1)^2}\right] \to 0\ (n\to\infty)$

$\theta = \frac{1}{1-\eta} - 1 \triangleq g(\eta)$ 在 $(0,1)$ 上连续，而 $X_i \in (0,1)$，$\bar{X} \in (0,1)$，$\eta \in (0,1)$，

故 $\hat{\theta} = \frac{1}{1-\bar{X}} - 1$ 是 θ 的相合估计.

三．(1). $p_X(x) = \frac{1}{\theta} e^{-\frac{1}{\theta}x}$，$x \geq 0$；$p_Y(y) = \frac{1}{3\theta} e^{-\frac{1}{3\theta}y}$，$y \geq 0$．由两样本相互独立，

$L(\theta) = \prod_{i=1}^n p_X(x_i) \prod_{j=1}^m p_Y(y_j) = \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta}\sum_{i=1}^n x_i} \cdot \left(\frac{1}{3\theta}\right)^m e^{-\frac{1}{3\theta}\sum_{j=1}^m y_j}$ —— 似然函数

取对数 + 求导 找最大值点

$\ln L(\theta) = -n\ln\theta - \frac{1}{\theta}\sum_{i=1}^n x_i - m\ln 3\theta - \frac{1}{3\theta}\sum_{j=1}^m y_j$

$\frac{d\ln L(\theta)}{d\theta} = \frac{-n}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} + \frac{-m}{\theta} + \frac{\sum_{j=1}^m y_j}{3\theta^2} = \frac{1}{3\theta^2}\left[3\sum_{i=1}^n x_i + \sum_{j=1}^m y_j - \theta(3n+3m)\right]$

$\frac{d\ln L(\theta)}{d\theta}$ 在 $\theta \in \left(0, \frac{3\sum_{i=1}^n x_i + \sum_{j=1}^m y_j}{3n+3m}\right)$ 上为正，在 $\theta \in \left(\frac{3\sum_{i=1}^n x_i + \sum_{j=1}^m y_j}{3n+3m}, +\infty\right)$ 上为负

$\therefore L(\theta)$ 与 $\ln L(\theta)$ 在 $\left(0, \frac{3\sum_{i=1}^n x_i + \sum_{j=1}^m y_j}{3n+3m}\right)$ 上单调递增．在 $\left(\frac{3\sum_{i=1}^n x_i + \sum_{j=1}^m y_j}{3n+3m}, +\infty\right)$ 单调递减.

$L(\theta)$ 最大值点：$\hat{\theta} = \frac{3\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j}{3n+3m}$ 是 θ 最大似然估计

(2). $E(X_i) = \theta$；$E(Y_j) = 3\theta$，$i = 1,2,\dots,n$；$j = 1,2,\dots,m$

$E(\hat{\theta}) = \frac{1}{3n+3m}[3\cdot n\cdot\theta + m\cdot 3\theta) = \theta$　是无偏估计

四. (1). 由题，$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$，$E(\bar{X})=\mu$，$\bar{X}$是$\mu$的无偏估计.

构造枢轴量：$\frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t(n-1)$

μ的$1-\alpha$置信区间：$-t_{1-\frac{\alpha}{2}}(n-1) \le \frac{\sqrt{n}(\bar{X}-\mu)}{S} \le t_{1-\frac{\alpha}{2}}(n-1)$

$$\Downarrow$$

$$\left[\bar{X}-\frac{S\, t_{1-\frac{\alpha}{2}}(n-1)}{\sqrt{n}},\ \bar{X}+\frac{S\cdot t_{1-\frac{\alpha}{2}}(n-1)}{\sqrt{n}}\right]$$

(2). $L=\frac{2}{\sqrt{n}}t_{1-\frac{\alpha}{2}}(n-1)\cdot S$，$L^2=\frac{4}{n}t^2_{1-\frac{\alpha}{2}}(n-1)\cdot S^2$

$D(L^2)=\frac{16}{n^2}t^4_{1-\frac{\alpha}{2}}(n-1)\,D(S^2)$.

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$. 故 $D\left[\frac{(n-1)S^2}{\sigma^2}\right]=2n-2$

凑$\chi^2(n-1)$求$D(S^2)$

$\therefore \frac{(n-1)^2}{\sigma^4}D(S^2)=2n-2$，$D(S^2)=\frac{2}{n-1}\sigma^4$

$\therefore D(L^2)=\frac{32}{n^2(n-1)}t^4_{1-\frac{\alpha}{2}}(n-1)\,\sigma^4$

五. 由题，$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$. $E(\bar{X})=\mu$，$\bar{X}$是μ的无偏估计

$D(\bar{X})=\frac{\sigma^2}{n}$. $p(x)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$，$\ln p(x)=-\frac{1}{2}\ln 2\pi\sigma^2-\frac{(x-\mu)^2}{2\sigma^2}$

$$\frac{\partial \ln p}{\partial \mu}=+\frac{x-\mu}{\sigma^2},\quad \frac{\partial^2 \ln p}{\partial \mu^2}=-\frac{1}{\sigma^2}$$

$I(\theta)=-E(-\frac{1}{\sigma^2})=\frac{1}{\sigma^2}$

$g(\mu)=\mu$，$g'(\mu)=1$　　方差下界：$\frac{[g'(\mu)]^2}{nI(\theta)}=\frac{\sigma^2}{n}=D(\bar{X})$，故$\bar{X}$是$\mu$的有效估计

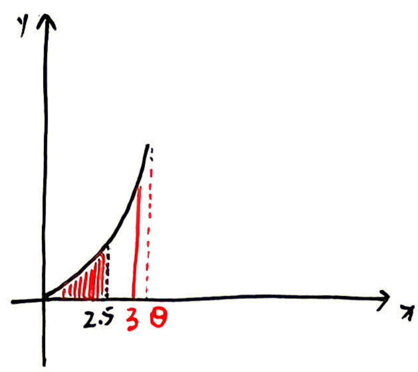
六. $X_{(n)} \sim p_n(x)=n\cdot\frac{x^{n-1}}{\theta^n}$，$x\in(0,\theta)$

$P(X_{(n)}\le 2.5 \mid \theta\ge 3)=\int_0^{2.5} n\cdot\frac{x^{n-1}}{\theta^n}dx=(\frac{5}{2\theta})^n$

当$\theta=3$时，$P(X_{(n)}\le 2.5\mid\theta\ge3)$取最大值. $\alpha=(\frac{5}{6})^n$

$(\frac{5}{6})^n\le 0.05$，$n\ge\frac{\ln 20}{\ln 6-\ln 5}\approx 16.64$

n至少取17



七. H_0：存活状况与性别无关. 拒绝域为 $W=\{\chi^2 \geq \chi^2_{0.95}(1)\}$

	男	女	
幸存	374	344	718
遇难	1364	126	1490
	1738	470	2208

$P_{11}=\frac{718}{2208}\times\frac{1738}{2208}$ ，$P_{12}=\frac{718}{2208}\times\frac{470}{2208}$

$P_{21}=\frac{1490}{2208}\times\frac{1738}{2208}$ ，$P_{22}=\frac{1490}{2208}\times\frac{470}{2208}$

$$\chi^2=\sum_{\substack{i=1,2\\ j=1,2}}\frac{(n_{ij}-n\cdot P_{ij})^2}{n\cdot P_{ij}}=\frac{n(n_{11}\cdot n_{22}-n_{21}\cdot n_{12})^2}{(n_{11}+n_{12})(n_{21}+n_{22})(n_{11}+n_{21})(n_{12}+n_{22})}$$

两种方法等价

$\chi^2\approx 450>3.8415=\chi^2(1)$ 故拒绝 H_0. 存活状况与性别有关.

八. (1).

来源	平方和	自由度	均方	F值
因子	~~[illegible]~~ 690	3	230	~~[illegible]~~ 2.0798
误差	4866	~~49~~	~~[illegible]~~ 110.59	
总和	~~[illegible]~~ 5556	47		

(2). ~~[illegible]~~ $2.0798 < F_{0.95}(3,44)=2.82$

因子A不显著

九. (1). $Q(a,b)=\sum_{i=1}^{8}(y_i-bx_i-a)^2$ ，$Q(\hat{a},\hat{b})=\min Q(a,b)$

$$\begin{cases}\frac{\partial Q}{\partial a}=-2\sum_{i=1}^{8}(y_i-bx_i-a)\\ \frac{\partial Q}{\partial b}=-2\sum_{i=1}^{8}(y_i-bx_i-a)x_i\end{cases}，\begin{cases}\frac{\partial Q}{\partial a}=0\\ \frac{\partial Q}{\partial b}=0\end{cases}\Rightarrow\begin{cases}na+n\bar{x}b=n\bar{y}\\ n\bar{x}a+\left[\sum_{i=1}^{8}x_i^2\right]b=\sum_{i=1}^{8}x_iy_i\end{cases}$$

由克拉默法则.

$$\therefore\ \hat{b}=\frac{\sum_{i=1}^{8}x_iy_i-n\bar{x}\bar{y}}{\sum_{i=1}^{8}x_i^2-n\bar{x}^2}，\quad \hat{a}=\bar{y}-\hat{b}\bar{x}$$

(2). $\hat{b}\approx\frac{15032-8\times24\times76}{4902-8\times24\times24}\approx1.4966$ ，$\hat{a}=\bar{y}-\hat{b}\bar{x}\approx40.0816$

$\hat{y}=40.0816+1.4966\hat{x}$

(3). $l_{xy}=\sum_{i=1}^{8}x_iy_i-n\bar{x}\bar{y}=440$. $l_{xx}=\sum_{i=1}^{8}x_i^2-n\bar{x}^2=294$ $l_{yy}=\sum_{i=1}^{8}y_i^2-n\bar{y}^2=886$

三种方法任选其一.

$$|r|=\frac{|l_{xy}|}{\sqrt{l_{xx}l_{yy}}}\approx0.8621>\sqrt{\frac{F_{0.95}(1,6)}{F_{0.95}(1,6)+6}}\approx0.71$$

显著

2024-2025 B

一. $\bar{X} \sim N(\mu_1, \frac{\sigma^2}{n})$, $\bar{Y} \sim N(\mu_2, \frac{\sigma^2}{n})$, 由于两样本相互独立,

$c(\bar{X}-\mu_1)+d(\bar{Y}-\mu_2) \sim N(0, (\frac{c^2}{n}+\frac{d^2}{m})\sigma^2)$, $\frac{c(\bar{X}-\mu_1)+d(\bar{Y}-\mu_2)}{\sqrt{(\frac{c^2}{n}+\frac{d^2}{m})\sigma^2}} \sim N(0,1)$

$\frac{(m+n-2)S_w^2}{\sigma^2} \sim \chi^2(n+m+2)$ 故 $\frac{c(\bar{X}-\mu_1)+d(\bar{Y}-\mu_2)}{\sqrt{(\frac{c^2}{n}+\frac{d^2}{m})\sigma^2}} \Big/ \sqrt{\frac{(m+n-2)S_w^2}{\sigma^2(m+n-2)}} \sim t(n+m-2)$

$\therefore \frac{c(\bar{X}-\mu_1)+d(\bar{Y}-\mu_2)}{S_w\sqrt{\frac{c^2}{n}+\frac{d^2}{m}}} \sim t(m+n-2)$

二. $\prod_{i=1}^{n} f(x_i,\lambda) = (2\lambda)^n (\prod_{i=1}^{n} x_i) e^{-\lambda(\sum_{i=1}^{n} x_i^2)}_{x\geq 0}$, $T(x_1,\dots,x_n)=\sum_{i=1}^{n} x_i^2$ 是λ的一个充分统计量,因为

$g(T,\lambda) = (2\lambda)^n e^{-\lambda(\sum_{i=1}^{n} x_i^2)}$, $h(x_1,\dots,x_n)=\prod_{i=1}^{n} x_i$, $\prod_{i=1}^{n} f(x_i,\lambda) = g(T,\lambda)\cdot h(x_1,\dots,x_n)$

三 (1). $E(X)=\int_{\theta}^{+\infty} 2\theta^2 x^{-2}dx = -\frac{2\theta^2}{x}\Big|_{\theta}^{+\infty} = 2\theta$. $2\hat{\theta}_1=\bar{x}$, $\hat{\theta}_1=\frac{1}{2}\bar{x}$ 是θ的矩估计

个人认为应该加个 $\theta>0$

$L(\theta)=\prod_{i=1}^{n} 2\theta^2 x_i^{-3} = \frac{(2\theta^2)^n}{\prod_{i=1}^{n} x_i^3}$ 关于θ在 $(0,+\infty)$ 单调递增

由于 $0<\theta \leq x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$, $\hat{\theta}_2 = x_{(1)}$ 是θ的最大似然估计

(2). 对 $\hat{\theta}_1=\frac{1}{2}\bar{X}$: $E(\hat{\theta}_1)=\frac{1}{2}\cdot\frac{1}{n}\cdot n\cdot E(X) = \frac{1}{2}\cdot\frac{1}{n}\cdot n\cdot 2\theta = \theta$. $\hat{\theta}_1$是θ的矩估计

对 $\hat{\theta}_2 = x_{(1)}$: $f(x,\theta)=\begin{cases} 2\theta^2 x^{-3} & x\geq\theta \\ 0 & x<\theta \end{cases}$ $F(x,\theta)=\begin{cases} 1-\frac{\theta^2}{x^2}, & x\geq\theta \\ 0 & x<\theta \end{cases}$

$x_{(1)} \sim p_X(x) = n\cdot(1-(1-\frac{\theta^2}{x^2}))^{n-1}\cdot\frac{2\theta^2}{x^3} = 2n\theta^{2n}\cdot\frac{1}{x^{2n+1}}$, $x\geq\theta$

$E(\hat{\theta}_2)=E(X_{(1)})=\int_{\theta}^{+\infty} 2n\theta^{2n}\cdot\frac{1}{x^{2n}}dx = \frac{2n\theta^{2n}}{1-2n}\cdot x^{1-2n}\Big|_{\theta}^{+\infty} = \frac{2n}{2n-1}\theta \neq \theta$. $\hat{\theta}_2$不是θ的矩估计

四. 构造枢轴量: $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

$P(\frac{(n-1)S^2}{\sigma^2} \geq a) = 0.95$, $a=\chi^2_{0.05}(n-1)$

$\therefore \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{0.05}(n-1)}$ 令 $n=9$, $\sigma^2 \leq \frac{8\times11\times11}{2.7326} \approx 354.34$. $\sigma \leq \sqrt{\frac{8\times11\times11}{2.7326}} \approx 18.82$

σ的单侧置信上限: 18.82

五、 $H_0: p_1=0.3 \quad p_2=0.3 \quad p_3=0.4$ $\qquad W:\{\chi^2 \mid \chi^2 \geq \chi^2_{0.95}(2)\}$

$$\chi^2=\frac{(56-200\times0.3)^2}{200\times0.3}+\frac{(69-200\times0.3)^2}{200\times0.3}+\frac{(75-200\times0.4)^2}{200\times0.4}=2.2625<2.7326=\chi^2_{0.95}(2)$$

∴ 与经验数据一致.

六、 $E(\bar{X})=\frac{1}{n}\cdot n\cdot E(X_1)=\theta$. $\bar{X}$是θ的无偏估计

$x\geq0$, $\ln f(x,\theta)=-\ln\theta-\frac{x}{\theta}$, $\frac{\partial}{\partial\theta}\ln f=-\frac{1}{\theta}+\frac{x}{\theta^2}$, $\frac{\partial^2}{\partial\theta^2}\ln f=\frac{1}{\theta^2}+\frac{-2x}{\theta^3}$

$$I(\theta)=-E\left(\frac{\partial^2}{\partial\theta^2}\ln f\right)=-\left(\frac{1}{\theta^2}-\frac{2\theta}{\theta^3}\right)=\frac{1}{\theta^2}$$

$g(\theta)=\theta$, $g'(\theta)=1$ $\qquad$ 方差下界: $\frac{[g'(\theta)]^2}{nI(\theta)}=\frac{\theta^2}{n}$

$D(\bar{X})=\frac{1}{n^2}\cdot n\cdot D(X)=\frac{1}{n}\cdot\theta^2=\frac{\theta^2}{n}$. 达到方差下界,故$\bar{X}$是$\theta$的有效估计

七、(1). 当$n=10$, $\mu=1$时, $\bar{X}\sim N(1,\frac{2}{5})$. $\bar{X}-1\sim N(0,\frac{2}{5})$. $\frac{\bar{X}-1}{\sqrt{2/5}}\sim N(0,1)$

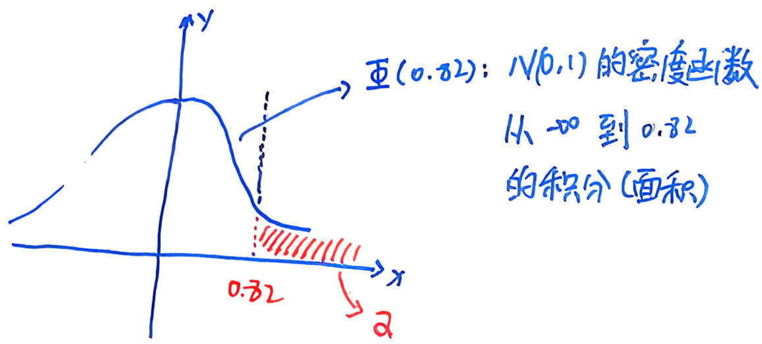
$\alpha=P(\bar{X}\geq1.52\mid\mu=1)$

$=P\left(\frac{\bar{X}-1}{\sqrt{2/5}}\geq\frac{0.52}{\sqrt{2/5}}\mid\mu=1\right)$

$=\left(1-\Phi\left(\frac{0.52}{\sqrt{2/5}}\right)\right)$

$\approx(1-\Phi(0.82))$

$=0.201$



(2). $\beta=P(\bar{X}\leq1.52\mid\mu=2)$ $\qquad$ $\mu=2$时, $\bar{X}\sim N(2,\frac{4}{n})$, $\frac{\bar{X}-2}{\sqrt{4/n}}\sim N(0,1)$

$=P\left(\frac{\bar{X}-2}{\sqrt{4/n}}\leq-\frac{6}{25}\sqrt{n}\mid\mu=2\right)$

$=\Phi\left(-\frac{6}{25}\sqrt{n}\right)\leq0.01$

$\Phi(2.32)=0.9808$

$\Phi(2.33)=0.9901$

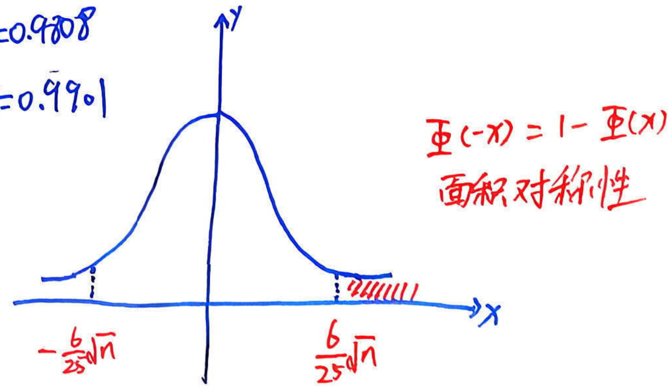
∴ $1-\Phi\left(\frac{6}{25}\sqrt{n}\right)\leq0.01$

$\Phi\left(\frac{6}{25}\sqrt{n}\right)\geq0.99$

试卷第一页

$2.33\leq\frac{6}{25}\sqrt{n}$

$n\geq94.25$



n至少取95

八. (1).

来源	平方和	自由度	均方	F值
因子	420	2	210	1.478
误差	3836	27	142.07	
总和	4256	29		

(2) $1.478 \leq F_{0.95}(2,27)=3.35$

无显著差异.

九. (1). $\hat{b}=\dfrac{\sum_{i=1}^{n}x_iy_i-n\bar{x}\bar{y}}{\sum_{i=1}^{n}x_i^2-n\bar{x}^2}$　　由 $y_i=a+bx_i+\varepsilon$, $\varepsilon\sim N(0,\sigma^2)$, $y_i\sim N(a+bx_i,\varepsilon)$

$$E(\hat{b})=\frac{\sum_{i=1}^{n}x_iE(y_i)-n\bar{x}E(\bar{y})}{\sum_{i=1}^{n}x_i^2-n\bar{x}^2}=\frac{\sum_{i=1}^{n}x_i(a+bx_i)-n\bar{x}(a+b\bar{x})}{\sum_{i=1}^{n}x_i^2-n\bar{x}^2}=\frac{na\bar{x}+b\sum_{i=1}^{n}x_i^2-na\bar{x}-nb\bar{x}^2}{\sum_{i=1}^{n}x_i^2-n\bar{x}^2}=b$$

$\therefore\hat{b}$ 是 b 的无偏估计

(2).

$$\hat{b}=\frac{\sum_{i=1}^{n}x_iy_i-n\bar{x}\bar{y}}{\sum_{i=1}^{n}x_i^2-n\bar{x}^2}=\frac{54107-12\times\frac{800}{12}\times\frac{811}{12}}{53418-12\times\frac{800}{12}\times\frac{800}{12}}\approx0.476$$

$\hat{a}=\bar{y}-\hat{b}\bar{x}\approx35.85$　　$\hat{y}=35.85+0.476\hat{x}$

(3).

$l_{xy}=\sum_{i=1}^{n}x_iy_i-n\bar{x}\bar{y}\approx40.33$　　$l_{xx}=\sum_{i=1}^{n}x_i^2-n\bar{x}^2\approx84.67$　$l_{yy}=\sum_{i=1}^{n}y_i^2-n\bar{y}^2\approx38.92$

$r=\dfrac{l_{xy}}{\sqrt{l_{xx}l_{yy}}}\approx0.70$　, $\sqrt{\dfrac{F_{0.95}(1,10)}{F_{0.95}(1,10)+10}}\approx0.576$　　$0.70>0.576$

线性关系显著